

Transcript: What's the problem?

Wilderness for Sale

Slide 2: Video Topics

- (Over)Population
- Carrying Capacity
- (Over)Consumption

This is the list of topics that will be covered in this video lecture. Population, or overpopulation, carrying capacity, and consumption or overconsumption.

Slide 3: (Over)Population

- World population growth
 - growth rate decreasing
 - the sum of the difference between births less deaths and the difference between immigrants less emigrants, for a country over one year (at the global level there is no migration)
 - fertility rates decreasing
 - average births per woman, > 2.1 is needed for growth
 - global population is still increasing, but more slowly
 - population growth is exponential

Near the turn of the nineteenth century, the world population reached a new milestone: one billion people. From there, it took more than a century to achieve two billion people and then far less time to triple that. Today, the world's population has surpassed eight billion people. And that's a whole lot of people.

So, what does the present look like and what does the future look like in terms of global population, growth, and decline?

The first thing I want you to know is that the population growth rate has been decreasing since it hit its peak in 1968. Today, the population growth rate is less than 1%. Keep in mind that this is the global rate of natural increase, births minus deaths since there is no migration at the global scale.

Also decreasing is the average number of births per woman. For a population to grow, this needs to be above 2.1, which is considered the replacement level. Below that, and at a global scale, the population will age and eventually decline. Many countries today have fertility rates below the replacement rate and extremely low rates of natural increase, but immigration keeps the populations of some of those countries growing.

Now let's talk about sheer numbers, because again, over 8 billion and growing means a lot of people. The one hundred years between 1920 and 2020 was a time of incredible growth and the global population nearly quadrupled.

Even with a slowing growth rate, population growth is exponential. The larger the base the greater the growth. Two people have two children, and those two children each have two children, and those four children each have two children, and on and on it goes.

Looking ahead to the end of this century, the global population is only expected to be about 25% larger, so somewhere around 10 billion in 2100, which is nowhere close to doubling.

The bottom line is that the global population is large, and increasing, but much more slowly than in the past.

Slide 4: Images "Comparison of United Nations population projections, Word" and "World population by region, including UN projections"

Population explosions are temporary, and we have seen this at the country level.

For many countries, that period of rapid growth has already ended and soon that will be the case for the world. The global fertility rate was cut in half between 1970 and 1990, which is a good indicator of where things are headed.

Up until 1700 – high childhood mortality counteracted high fertility and the population grew very slowly.

The Industrial Revolution happened, the Medical Revolution happened, health improved and mortality declined. At that point, things changed quickly, first in the early-industrialized countries and then spreading to the rest of the world. Between about 1950 or so and 2020, which is the steepest part of that graph on the right, the global population more than tripled, with a significant number of people—more than 3.5 billion—being added in the latter half of the twentieth century.

In the two centuries between 1800 and 2000, there was a 7-fold increase in population and this is when the greatest environmental change took place.

Technological advances coupled with explosive population growth amplified humanity's impact on the natural environment. In the 20th century we saw rivers burn, oil spill, smog kill, and black rain. The environment was sounding an alarm.

Providing space, food, and resources for an incredibly large population for decades to come, while also minimizing or mitigating damage, is one of the biggest challenges we face today.

We are still adding tens of millions of people to the population every year and we must find a way to provide for them without completely disabling the environment's ability to do so.

The United Nations projects that the global demographic transition will end, population growth will continue its decline and the curve representing the world population will level off. It is expected that the population growth rate will fall to 0.1% and from there, who knows? The population projection, the medium variant, is just a little more than 10 billion people in the 2080s before it levels off and then declines.

A lot of this is dependent on fertility rates, and fertility rates decrease when economic and human development increase.

You can see in the graph on the right that most of the greatest growth moving forward will happen in countries of Africa and Asia, however, Asia, with an incredibly large base population, has seen some pretty significant decreases in fertility rates.

Slide 5: (Over)Population

- Characteristics of growth today
 - population growth (and fertility rates) is highest in the least developed countries (UN Classification)
 - population growth rate is negative in some countries (notably Japan and Germany)
 - in some countries, immigration accounts for more than half of population growth

Keeping with that thought, the world fertility rate is falling, but it remains high in many developing countries. According to the United Nations, just eight countries will account for more than half of the world's population growth between now and 2050, five in Africa and three in Asia.

This will put a huge strain on resources in those countries, not just necessities like food and water, but housing,

schools, and utilities as well. Plus, government programs and policies that focus on things like reducing poverty, improving healthcare, and sustainability, face even greater pressure.

Life expectancy has increased, and infant mortality has decreased, but it is fertility and family size that are driving growth in most countries.

The base population is also a factor, India and Nigeria are two of the most populous countries on the planet. Nigeria's fertility rate remains exceptionally high, while India's has fallen below replacement level. Even if fertility rates were to fall, as they have in India, it takes decades before that is reflected in the population.

Other countries are experiencing negative population growth or decline. Japan is one country whose population has been on the decline for some time now. China is facing a similar situation, and many other countries have stable, declining, or soon-to-be declining numbers as well.

In Canada, immigrants account for almost three-quarters of population growth, and in the United States, even more than that.

Slide 6: Images of population growth, food security, fertility and poverty

Our World in Data is a great resource for charts, visualizations, maps, and online or embedded, they are also interactive. And you'll notice I use them quite a bit.

These maps are not current, but they do paint a picture. Population growth has slowed dramatically in many countries, but it is still high in a handful of countries, and many of those countries will account for much of the growth in the decades to come.

Many countries with growing populations are considered least developed by the UN classification system, and if you are unsure what that entails, I encourage you to look it up.

Some of the countries with the fastest-growing populations are also those with the highest poverty rates, and many of those living in poverty are found in rural areas. It is estimated that 80% of the world's poor live in rural areas and the poverty rate in rural areas is more than three times higher than it is in cities.

These people are those most dependent on the environment for their subsistence needs and livelihood. They are heavily reliant on ecosystem services, highly vulnerable to the negative impacts associated with climate change, and those least able to do something about it.

Now, that is not to say that more developed countries don't have poverty or food insecurity issues. And really, you could argue that in a high-income or rich nation, poverty and food insecurity should be close to nothing, but of course, that is not the case.

Slide 7: (Over)Population

- Thoughts on growth & the environment
 - Neo-Malthusianism
 - uncontrolled population growth will lead to ecological collapse and must be curbed
 - Cornucopianism
 - technology and human ingenuity will save us
 - Conflict Theory
 - humans are in competition for resources, the issue is not a lack of resources but the inequitable distribution of those resources

Now I am going to transition into a conversation about population size and resource use. When the population is viewed as a factor of production, people are labor and more labor equals more production. That is assuming

that people, as laborers, are used efficiently. More labor also means more talent to develop new ideas and advance society.

The flip side to this is that a large population also puts a strain on finite resources.

And this tension has long been a debate for economists.

The debate between ecologists, environmental economists, sociologists, and population geographers, usually takes a more functional approach considering the complex relationship between humans and the environment, and this is what I want to consider here.

So, first, we have the Neo-Malthusian view, and I will get to who Thomas Malthus is in just a moment, but Neo-Malthusians believe that left uncurbed, population will lead to ecological collapse. They are strong proponents of birth control and coercive population policies, which are policies that take away a woman's or family's choice on the number of children they will have through a system of legal or economic penalties and rewards.

Technology might be the greatest opposition to the idea that the world is limited in the number of people it can support without collapse. Humans have an enormous capacity to enhance or weaken the productivity of the environment and tools, technologies, and organizations, and using resources efficiently can increase the number of people that can be supported by a resource base.

This is the idea behind Cornucopianism and those who believe that human ingenuity is capable of solving all challenges to growth and that population growth is a catalyst for greater ingenuity.

Last but not least, is conflict theory, which isn't typically included in the debate between Cornucopians and Neo-Malthusians, but the idea behind conflict theory is that humans have an internal drive to survive and compete, and the resources exist to support the population.

But the human population complicates this with economic and political systems that make for an unfavorable balance. So it's not the lack of resources, but the inequitable and inefficient distribution and use of those resources that create the problem. And there is some evidence to support this.

Slide 8: Image showing a boat with a quote from Garrett Hardin

Garrett Hardin was an ecologist and white nationalist, born in Texas, and he most notably authored the essay "The Tragedy of the Commons" which we will talk about later, but he also wrote a ton of books and articles. He was majorly opposed to socialism and welfare, compassion really wasn't his thing I guess, and on this slide is a quote from a journal article that he wrote. It really sums up the hardline of the Neo-Malthusian view.

Essentially, if you are rich and in a lifeboat surrounded by the world's poorest people, and you have room for a few more people, what should you do?

If you let them all in, the boat sinks and everyone dies.

If you let some in, who do you let in? And the bigger issue, for Hardin, was that the boat is now at its limit, which puts everyone at risk.

Hardin felt that the answer was obvious, the rich should float and the poor should be left to sink or swim.

This is obviously an extreme example from someone who was a racist and a fascist, but also very neo-Malthusian.

Slide 9: Image "Change in corn production and land use in the United States"

And we have images like this that help bolster the Cornucopians' view that human ingenuity and technology can figure this out. In the United States, as an example, corn production skyrocketed while cultivated land remained largely unchanged.

The Green Revolution was a program that relied on agricultural innovation and biotechnology to greatly increase the world's food supply. While the world's population was growing, technology was also advancing to increase agricultural productivity and keep pace with population growth.

Of course, the Green Revolution also put incredible strain on the environment and upended traditional farm systems, but that is not a conversation for this lecture.

Slide 10: Carrying Capacity

- Ecosystems are defined by their most limiting factor
- Thomas Malthus
 - humans need food, and
 - humans have the desire to engage in sexual relations (and, thus, procreate).
 - if there is not enough food, misery, and vice will result

People living on Earth rely on resources that the planet supplies, many of which are finite. Carrying capacity is the idea that the physical environment determines the number of creatures that can be sustained indefinitely by an ecosystem or resource base.

It is an idea that was first formulated by herders and pasture managers to determine the largest number of cattle or sheep they could stock on their range without degrading the soil or vegetation. It's really a common sense idea that if a species is dependent on a number of things to live and one of those things is deficient, it does not matter how abundant the other components are, the species will not survive. If a plant depends on water, soil nutrients, and sunlight to live, and water is withheld, the plant will die.

The carrying capacity of any system is always defined by its most limiting component—not its most abundant. For humans, the most influential variable associated with carrying capacity is climate. This doesn't mean that climate is the limiting factor, rather that climate, at this point in history, exerts the biggest influence on the availability of food and water.

In 1798, Thomas Robert Malthus, a clergyman, and mathematician by trade was one of the first to publish his opinion of Earth's most limiting factor regarding its inhabitants. Malthus felt quite strongly that the yield of arable or farmable land would be what ultimately limited the number of people the planet could support. In other words, food.

Malthus was really a pioneer in thinking about population from a resource perspective because *at this time*, people were pretty much viewed only as a resource, and for that many people called him a pessimist. He also failed to predict any of the technological advances and his formulation was a bit off, but there is a hint of validity here.

Slide 11: Image and Map of Sahel & Senegal

An excellent example of carrying capacity and people-environment interactions can be seen in the region of Africa called the Sahel. This region lies on the southern fringe of the Sahara and is bordered by tropical rainforests in the south. It is a place of highly variable environmental conditions, ranging from wet to dry, both on a seasonal and annual basis. For millennia, pastoralists respected the Sahel's fragile natural environment, migrating their herds to follow water and avoid overgrazing.

But, humans have an enormous capacity to enhance or weaken the productivity of the environment; tools, technologies, organization, and using resources efficiently can increase the number of people that can be carried on a resource base. The maximum population that can be supported by a region is continually

changing due to the actions of humans. It is impossible to accurately predict what the earth's carrying capacity is, but it can never grow larger than subsistence will allow without consequence.

Colonization by European Powers is a chapter in the story of the African continent, and it covered a time of great change. Non-native people occupied the Sahel during that time and they imposed many new rules and boundaries that were not established with the local climate in mind. Wildlife, people, and livestock were restricted and unable to move freely across the Sahel. Life became more sedentary and fewer people relied on agriculture and pastoralism to make their living. Many were forced to take part in the European's exploitation of the region's natural resources. With a more sedentary lifestyle also came localized human and livestock population growth and urbanization.

In other words, many more people were living in a more concentrated area.

A number of factors, increasing population growth, increasing urbanization, diminishing water resources, a lack of mobility due to land ownership and political boundaries, more intensive soil use, the expansion of commercial agriculture, overgrazing, livestock overpopulation due to vaccinations, and a lack of government support for pastoralism because the profession was considered by some to be an obstacle to modernization...all of these things contributed to one outcome: the Sahel's carrying capacity has been far exceeded.

And this outcome means that any time there is a social or environmental stressor, like a conflict or period of drought, the result is catastrophic for humans and the environment. The Sahelian environment has become stressed, consequently, degraded.

Slide 12: (Over) Consumption

- Some thoughts on consumption
 - People do not consume equally
 - The United States accounts for 5% of the world's population and
 - 25% of the world's energy consumption
 - produces nearly 33% of the world's hazardous waste
 - consumes more calories and water per capita on a daily basis
 - Is it easier to impose a population policy than a consumption policy?

A really small portion of the global population uses most of its resources and produces most of its greenhouse gas emissions, at least from a cumulative standpoint.

Since the dawn of the Industrial Revolution, the United States and the countries of the European Union account for roughly half of the total, cumulative, greenhouse gas emissions, and the United States alone represents about 30% of that. It's only in the last several decades that countries of Asia, Africa, and South America, have seen sizable increases in emissions, but their relative, cumulative contribution remains low, with the exception of China.

China and India outrank the United States in terms of annual freshwater consumption, but when viewed per capita, the story changes and water use per person is significantly higher in the United States. Greater industrial and municipal use contribute to this; agriculture is pretty much a huge water suck worldwide.

A number like 8 billion or 9 billion is a great scapegoat for environmental degradation, and policies have tended to focus on getting that in check, but without examining patterns and levels of consumption, would it make a difference if there were only 2 billion people in the world, but they all consumed as much as the average American?

I am picking on America because it is the largest consumer, with the largest economy, and the greatest wealth, as well as the country where I live and am most familiar with, but excessive consumption is an issue wherever it takes place. Also, the least developed countries tend to get a bad rap for high population growth. The bottom

line is that In most cases, resource consumption is more a reflection of industrialization and affluence, or standard of living, than the sheer size of a population.

Charles Kenny, an economist and Senior Fellow at the Center for Global Development in Washington, has been quoted as saying “Population is not the problem, the way we consume is the problem — let’s change our consumption patterns.” The environmental concerns surrounding the 8 billion mark, he feels, should shift their focus to consumption, mostly in developed countries.

Slide 13: Concluding Thoughts

- How can we use available resources (land and water) to produce food for all?
- How can we promote economic development and end poverty so that all can afford to eat?
- How should we respond to the human and environmental consequences of industrialization?

This is a human and the environment class, and a good approach to evaluating the global (over-)population problem is from that of a human-environment perspective. The potential overpopulation problem does not, should not, rest solely on the global population.

Rather, the problem rests on the relationship between the number of people (population) and the availability of natural resources (environment).

It is when the population of a given area exceeds the capacity of the surrounding environment to support it at an acceptable standard of living that overpopulation problems arise. And, as you probably suspect, such overpopulation problems either are or are threatening to be, major in some regions of the world and much less of a problem in others. Some regions have a favorable balance between the number of people living there and the available resources, while others do not.

When thinking about how we should approach this problem, the United Nations notes that limiting family size through removing the freedom of choice is not the preferred or most effective path to curbing population growth, but empowering women, increasing educational opportunities, and offering them reproductive options not only reduces population growth but also promotes environmental conservation.

The United Nations Sustainable Development Goals are one way the world is coming together to help bring greater balance between people and the environment. They note that as globalization has increased, it has amplified uneven development, with prosperity in some locations and poverty in others.

Smart policymakers are considering questions like those I’ve included here: how can the resources available provide for the population, how can economic development help end poverty and end food insecurity for everyone, and how should we address the negative outcomes of industrialization including global climate change, degrading air, water, and soil, and the loss of biological diversity.

And with that, I will wrap up this lecture on Population and Consumption.